27th National High School Physics Finals: Solutions

I. (25 points) Fill in the blanks

1. $\dfrac{4\pi^2 kR}{4\pi^2 k - M\omega^2}$ （6 points）

2. $\dfrac{fV}{V+u}$, $\dfrac{fV}{V-u}$ （4 points）

3. $T_0\left(1-\dfrac{Mg}{p_0 A}\right)^{\frac{1-1}{\gamma}}$ （6 points）

4. $7.3\times10^6\,\mathrm{m\cdot s^{-1}}$, $2.8\times10^8\,\mathrm{m\cdot s^{-1}}$, $8.4\times10^5\,\mathrm{V}$ （9 points）

$$p=\frac{h}{\lambda}$$

$$E=\sqrt{p^2c^2+m_e^2y}$$

$$E_k=eV=E-m_ec^2$$

II. (20 points)

1. Let m be the block mass. Its acceleration down the incline is

$$a=\frac{mg\sin\theta-\mu_1 mg\cos\theta}{m}=g\left(\sin\theta-\mu_1\cos\theta\right),\qquad(1)$$

The speed of the block at the bottom of the incline is

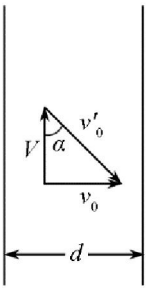
$$v_0=\sqrt{\frac{2ah}{\sin\theta}}=\sqrt{2gh\left(1-\mu_1\cot\theta\right)}=4.0\,\mathrm{m/s}\qquad(2)$$

In the belt frame, the initial speed of the block entering the belt is

$$v_0'=\sqrt{v_0^2+V^2}=5.0\,\mathrm{m/s}\text{。}\qquad(3)$$

The angle α between the motion and the belt edge satisfies

$$\tan\alpha=\frac{4}{3}\text{。}\qquad(4)$$



The block decelerates on the belt with acceleration magnitude

$$a'=\frac{\mu_2 mg}{m}=\mu_2 g\text{。}\qquad(5)$$

The distance traveled relative to the belt before coming to rest is

$$s'=\frac{v_0'^2}{2a}=\frac{v_0'^2}{2\mu_1 g},\qquad(6)$$

The least friction coefficient μ_2 preventing escape over the belt edge satisfies